

“ ”

“ XLS web GUI
GUI

“ ” ”

XLS web

81

id	ID
----	----

Project

TeamExp

ManagerExp

YearEnd

Length

Effort

Transactions

Entities

PointsAdjust

0) GUI 81, 4

600px|

$$\begin{array}{ccc} 1 & & 4 & -1 \end{array}$$

600px|

600px| 2

Box-cox

 $\log$

600px|

3 (correlation) 2

600px|

4) 2

$$1\ 2 \qquad 3 \qquad 1\ 2$$

600px|

600px|

600px|

Effort (Length) (Effort) Y
5) LogEffort XLS RMSE R-square
600px|

RMSE (Root Mean Square Error) R-
square R-square 1 2 0 2 (RMSE R-
square)

: X Y $y = a + bx$ 3 X
 $y = a + bx_1 + cx_2 + dx_3$ ElasticNet
Cubist

-
- 2
1. PtsAdjust + TeamExp
 2. PtsAdjust + TeamExp + MgrExp
 3. PtsAdjust + TeamExp + MgrExp + Language
- Language 2

650px

• (#1) (#2), RF
• RF 2
PCR ElasticNet
(Random Forest) (overfitting) “ ”
X TeamExp LogPtsAdjust 0.27 TeamExp LogEffort 0.22 LogEffort LogPtsA
2 10-Fold, Cross Validation:
10

R-square RMSE R-square
650px

2
excel XLS 10-Fold Cross Validation

R R
R 20% 10% R
(package)

R tidyverse
R tidyverse R 5 R R
AI AI

R tidymodel parsnip random forest machine learning via ranger R code example

AI ranger

“ ”

“ ”

“ ”

“ ”

AI AI X c 123 ”

“ ”

“ bug

PAMAS 2009 ‘ , ’ ”

References

1. (dataset) : <https://www.kaggle.com/datasets/toniesteves/desharnais-dataset>
2. Philippe J.S. De Brouwer *The Big R-Book* (Wiley , 2021)
3. Kuhn, Max and J. Silge *Tidy Modeling with R : A Framework for Modeling in the Tidyverse.* (O'Reilly , 2022)

Category:2410Edition